C. Balkı GEMİRTER

Chief AI Officer (CAIO) & R&D Leadership

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SUMMARY

Ph.D. candidate in Computer Engineering at Yeditepe University (4.00 GPA) with 19+ years of enterprise R&D leadership and deep-tech entrepreneurship experience. Built upon a strong corporate engineering foundation at TEB and Nortel/Netaş, with concurrent leadership in 0-to-1 AI architectures, TÜBİTAK-funded projects, and LLM implementations as CAIO and AI Director across high-growth startups (CEIBA Health, Bayegan, Agada Tech, Vivoo). Specialized in transforming advanced AI research into scalable products and measurable business value in healthcare, finance, and commodity trading.

SKILLS

AI & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Generative AI (GenAI), Large Language Models (LLM), Natural Language Processing (NLP), Computer Vision, Question Answering (MRQA), BERT, YOLO, GANs, Anomaly Detection, Fraud Detection, Predictive Analytics, Customer Segmentation, Domain Adaptation, Signal Processing (DSP)

Tools & Technologies: PyTorch, TensorFlow, Apache Spark, Java, Oracle, AMQP, MQTT

Leadership & Management: AI Strategy & Roadmapping, R&D Management, TÜBİTAK TEYDEB 1501 / 1832, EU Horizon 2020, Grant & Incentive Management, Team Leadership (up to 17 engineers), Scrum (PSM I, PSPO I), Product Ownership

Industries: HealthTech, FinTech & Banking, Commodity Trading, E-commerce, Telecommunications, IoT & Smart Cities

WORK EXPERIENCE

Ph.D. Candidate · Teaching Assistant & Guest Lecturer

Yeditepe University | Sep 2021 - Present

- Doctoral research on incorporating prior knowledge into deep learning architectures for image recognition and medical diagnosis with small cohort data (expected graduation: December 2026)
- Teaching assistant for Artificial Intelligence (CSE 462), Machine Learning (CSE 585) and Deep Learning (CSE 588)
- Designs and grades the term projects and assignments of these courses
- Guest lecturer for the university-wide AI 401 course and at HSE University (Moscow)

Director of Artificial Intelligence

CEIBA HEALTH | Jan 2026 - Apr 2026

- Led AI and R&D strategy for next-generation healthcare solutions
- Secured and managed TÜBİTAK 1501 and 1832 projects
- Directed YOLO-based real-time fall risk detection (>90% accuracy, <10s alert latency) and architected both the LLM-powered "AI Companion" for automated SOAP notes and a DSP-based strip analysis system for ECG rhythm analysis

Chief AI Officer (CAIO)

Bayegan | May 2025 - Jul 2025

- Spearheaded the AI roadmap of a commodity trading company, aligning it with business objectives to enhance predictive analytics and risk management
- Directed model training and optimization for automated trading systems
- Fostered collaboration with trading desks and software engineers to integrate AI into workflows

VP of Artificial Intelligence

Agada Tech | Sep 2024 - Jan 2025

- Spearheaded AI-driven customer segmentation models and oversaw real-time next-order prediction systems for high-volume e-commerce

- Pioneered advanced AI methodologies to enhance customer loyalty strategies

AI & Applied Data Science Manager
Yapı Kredi Teknoloji | May 2024 - Aug 2024
- Led a team of 17 engineers with 0% attrition
- Managed critical AI projects such as Digital Mizan and Fraud Detection
- Delivered cost and FTE savings through computer vision, anomaly detection, and deep learning
- Led ITU Technopark-supported R&D and contributions to Horizon 2020

Chief AI Officer (CAIO) & Data Analytics
Vivoo | Aug 2023 - Feb 2024
- Managed the development of "Welly", an LLM-based wellness assistant, and AI-driven test strip analysis tools
- Led a TÜBİTAK 1501-supported project analyzing nutritional values from meal photos

Senior R&D Manager (Artificial Intelligence, Software)
TEB - Türk Ekonomi Bankası | Aug 2016 - Aug 2023
- Led a team of 10 engineers in TEB's R&D department
- Idea owner and supervisor of the TÜBİTAK TEYDEB 1501-supported NEURON (AI chatbot) and ConnecTEB projects
- Led deep learning/NLP initiatives such as Dragon and Ribbon
- Pioneered the transition of the first R&D team to TEB ARF

Infrastructure Software Development Assistant Project Leader
TEB - Türk Ekonomi Bankası | Oct 2015 - Aug 2016
- Architect of the core IT framework used by TEB's software designers across all banking projects
- Led scalability, performance, and new-requirement integration efforts

Switch and ATM Software Development Assistant Project Leader
TEB - Türk Ekonomi Bankası | Nov 2014 - Oct 2015
- Core member of the Aquila team on TEB's first TÜBİTAK TEYDEB 1501 project, a 100% in-house XTM system
- Led a team of 5 engineers from prototype to full-scale deployment

Internet Banking Software Development Project Supervisor
TEB - Türk Ekonomi Bankası | Jan 2013 - Nov 2014
- Team lead for over 10 engineers across retail and commercial internet banking
- Spearheaded the redesign of both platforms and delivered an advanced smart search feature

Senior Software Design Engineer
NORTEL Networks - NETAŞ | Mar 2010 - Jan 2013
- Designed and developed the A2 Base/OAM system manager and its sub-components using Java and Oracle
- Built high-performance, resilient architectures for critical network infrastructure

Junior Software Design Engineer
NORTEL Networks - NETAŞ | Jul 2007 - Mar 2010
- Developed and tested system management (OAM) software components in Java for telecommunications network infrastructure

EDUCATION

Ph.D. - Computer Engineering
Yeditepe University | Taught in English | 2020 - Dec 2026 (Expected)
GPA: 4.00/4.00 (Ranked 1st, Full Scholarship).
Thesis: Incorporating prior knowledge in Deep Learning (DL) architectures for image recognition and medical diagnosis with small cohort data.

M.Sc. - Computer Engineering
Yeditepe University | Taught in English | 2018 - 2020
GPA: 3.87/4.00 (Ranked 1st).
Thesis: A Turkish Question Answering System Based on Deep Learning Networks (BERT).

B.Sc. - Computer Engineering
Ege University | 2002 - 2007
GPA: 3.34/4.00 (Ranked 4th). Education in core computer engineering disciplines and software architecture.

ALES: 88 (2018) - Academic Personnel and Postgraduate Education Entrance Exam
YÖKDİL: 85 (2026) - Higher Education Institutions Foreign Language Exam (English)

PUBLICATIONS

- **RP-Fuse: Development of an Adaptive Dose-Gated Multimodal Model for Predicting Radiation Pneumonitis in Non-Small Cell Lung Cancer**. C. Balkı GEMİRTER, Esra KORKMAZ KIRAKLI, Cemre Ceren ÖZDİLLİ, Ali Kadri ÇIRAK, E. Erkan KORKMAZ, Dionysis GOULARAS. *MDPI - Applied Sciences (2026)*. SCIE - Q2.
- **CoDA: A Cognitive-Inspired Approach for Domain Adaptation**. C. Balkı GEMİRTER, E. Erkan KORKMAZ, Dionysis GOULARAS. *MDPI - Applied Sciences (2026)*. SCIE - Q2.
- **Location Proofing Using Video Similarity**. C. Balkı GEMİRTER, Tacha SERIF. *Journal of Scientific Reports-A (2022)*. TR Dizin.
- **A Turkish Question Answering System Based on Deep Learning Neural Networks**. C. Balkı GEMİRTER, Dionysis GOULARAS. *JISTA - Journal of Intelligent Systems (2021)*. TR Dizin.
- **Fine-Tuning Throughput and QoS on SMT Cores**. C. Balkı GEMİRTER, Gürhan KÜÇÜK. *UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021)*. IEEE.
- **A Comparative Evaluation of AMQP, MQTT and HTTP Protocols Using Real-Time Public Smart City Data**. C. Balkı GEMİRTER, Çağatay ŞENTÜRCA, Şebnem BAYDERE. *UBMK 2021 - 6th International Conference on Computer Science and Engineering (2021)*. IEEE.

R&D PROJECTS

Image Processing-Supported Patient Monitoring and Fall Risk Management
CEIBA HEALTH | TÜBİTAK TEYDEB 1501 | 2026
Supervised R&D project focusing on AI-based healthcare solutions.

Green and Sustainable Digital Solutions in the Health Sector
CEIBA HEALTH | TÜBİTAK TEYDEB 1832 | 2026
Supervised R&D initiative for eco-friendly digital health.

AI-Based Dynamic Customer Segmentation and Personalized Gamification System
Agada Tech | TÜBİTAK TEYDEB 1501 | 2024 – 2025
Advanced AI models for user retention and personalized experiences.

AI-Based Nutrition Analysis System
Vivoo | TÜBİTAK TEYDEB 1501 | 2023 – 2024
Multimodal health AI project expanding personalized nutrition insights.

Dragon - Banking Question Answering (MRQA) System
TEB - Türk Ekonomi Bankası | Boğaziçi Technopark | 2022 – 2023
Machine reading comprehension-based smart search over banking documents, giving call center staff instant, concise answers to banking queries.

AI-Based Customer Chatbot
TEB - Türk Ekonomi Bankası | TÜBİTAK TEYDEB 1501 | 2020 – 2022
Developed NEURON, an AI chatbot integrated into digital banking.

ConnecTEB - VoIP & Multimedia Solution
TEB - Türk Ekonomi Bankası | TÜBİTAK TEYDEB 1501 | 2016 – 2018
In-house secure video conferencing tool adopted bank-wide.

Orion & Aquila (In-House XTM)
TEB - Türk Ekonomi Bankası | TÜBİTAK TEYDEB 1501 | 2014 – 2016
Core IT framework development and scaling of ATM/Switch infrastructure.

InVisIBLe Plus
Yeditepe University | EU-Supported
Inclusive and innovative learning tool for visually impaired and blind people.

ACADEMIC & PROFESSIONAL SERVICE

- **Reviewer**, Turkish Journal of Electrical Engineering & Computer Sciences | TÜBİTAK
- **Reviewer**, European Journal of Radiology | Elsevier
- **Mentor**, Bilişim Vadisi (Informatics Valley) | Technopark
- **Guest Lecturer**, HSE University | Spring 2025 & 2026
  Invited 3-hour Artificial Intelligence lectures at HSE University (Moscow) in two spring terms; designed the course term projects.
- **Teaching Assistant**, Yeditepe University | 2021 – Present
  Teaching assistant to Prof. Dr. E. Erkan Korkmaz for Artificial Intelligence (CSE 462, AI 401), Machine Learning (CSE 585) and Deep Learning (CSE 588), and to Dr. Onur Demir for Algorithm Analysis (CSE 311); designed and graded term projects and assignments.

CERTIFICATIONS

- Generative AI for Everyone - DeepLearning.AI (Mar 2025)
- AI for Medicine Specialization - DeepLearning.AI (Jul 2022)
- Sentiment Analysis with Deep Learning using BERT - Coursera (Feb 2022)
- Scalable Machine Learning on Big Data using Apache Spark - IBM (Nov 2021)
- Deep Neural Networks with PyTorch - IBM (Aug 2021)
- Generative Adversarial Networks (GANs) Specialization - DeepLearning.AI (Feb 2021)
- Natural Language Processing Specialization - DeepLearning.AI (Dec 2020)
- Fundamentals of Parallelism on Intel Architecture - Intel (Nov 2020)
- Machine Learning - Stanford Online (May 2020)
- Deep Learning - DeepLearning.AI (Apr 2020)
- TensorFlow in Practice Specialization - DeepLearning.AI (Mar 2020)
- PSPO-I Professional Scrum Product Owner - Scrum.org (Mar 2020)
- PSM-I Professional Scrum Master - Scrum.org (Feb 2017)

AWARDS

- Leaders of Tomorrow (x2) - Global Talent Award (2023 - 2024)

LANGUAGES

Turkish (Native), English (Professional), Japanese (Elementary)